University of Asia Pacific

Department of Computer Science & Engineering

CSE 414

Machine Learning & Deep Learning Lab

Project Proposal

of

Multi-Class Sentiment Analysis of Bangla Social Media Comments using Machine Learning & Deep Learning

Submitted by

Marzan Ahmed (ID: **22201055**)

Md. Sahriar Asif (ID: **22201111**)

Faiza Ahmed Esha (ID: **22201121**)

Lubaba Hasan (ID: **22201134**)

Section: B

Submitted to

Faria Zarin Subah

Assistant Professor,

Dept. of CSE, UAP

Date of Submission: April 18, 2026

Table of Contents

Project Title 2
Abstract 2
1. Introduction 2
    1.1 Problem Statement 2
    1.2 Motivation 2
    1.3 Objectives 2
2. Dataset 3
3. Methodology 3
    3.1 Data Preprocessing 3
    3.2 Feature Extraction 3
    3.3 Model Implementation 4
    3.4 Training 4
    3.5 Implementation Tools 4
4. Evaluation & Ablation Study 5
    4.1 Evaluation Metrics 5
    4.2 Validation Strategy 5
    4.3 Ablation Study 5
5. Gantt Chart 6
6. Expected Outcome 6

Project Title

Multi-Class Sentiment Analysis of Bangla Social Media Comments using Machine Learning & Deep Learning.

Abstract

Sentiment analysis is a key task in natural language processing (NLP) for understanding public opinion. While significant progress has been made for English, Bangla remains underrepresented due to limited high-quality datasets and linguistic challenges. This project aims to develop a machine learning-based sentiment analysis system for Bangla social media comments using a fully self-collected and manually labeled dataset. The model will classify comments into explicitly toxic, subtly toxic and neutral categories. By focusing on real-world, noisy data and building a dataset from scratch, this project contributes both a practical system and a valuable resource for Bangla NLP domain. The results of this research can be useful for applications such as opinion mining, brand monitoring and social trend analysis.

1. Introduction

1.1 Problem Statement

Existing sentiment analysis models largely focus on English and rely on well-structured datasets. For Bangla, there is a lack of high-quality, diverse and representative datasets, especially for informal social media text. This makes it difficult to build accurate sentiment classification systems.

1.2 Motivation

The rapid growth of Bangla content on social media platforms such as Facebook, YouTube and Twitter has created a strong demand for automated sentiment analysis tools. However, there is a shortage of reliable sentiment analysis tools tailored specifically for Bangla. Developing such systems is therefore important, as they can help businesses, policymakers and researchers better understand public opinion, user behavior and social trends expressed in Bangla online content.

1.3 Objectives

- Develop a sentiment analysis model trained on pure Bangla text data.
- Utilize a self-collected and manually labeled dataset as the primary data source.
- Supplement the dataset with external data to improve diversity and generalization.
- Evaluate multiple machine learning approaches on this combined dataset.

Unlike many existing works that rely on pre-existing datasets, this project emphasizes data creation as a core contribution. A well-curated dataset combined with a strong

baseline model can serve as a foundation for future Bangla NLP research and applications.

2. Dataset

The dataset used in this project is a combination of **self-collected** Bangla social media comments from social media platforms such as Facebook and YouTube and externally collected datasets from sources such as **Kaggle**.

Dataset Size: 20,000 – 30,000 samples.

3. Methodology

The proposed system is designed as a supervised multi-class classification model for Bangla sentiment analysis. The overall pipeline consists of data preprocessing, feature extraction, model implementation and training.

Comments with less than 40% Bangla Unicode characters (U+0980–U+09FF) are removed, ensuring the model trains on Bangla text only.

3.1 Data Preprocessing

Raw social media comments are inherently noisy and require cleaning before model training. The following preprocessing steps are applied:

- **Language filtering** based on comments with less than 40% Bangla Unicode characters (`U+0980–U+09FF`) are removed, ensuring the model trains on Bangla text only.
- **Removal of noise** such as URLs, emojis and special characters.
- **Elimination of Bangla stopwords** to reduce redundancy.
- **Tokenization** of text into meaningful units suitable for analysis.

3.2 Feature Extraction

To convert textual data into numerical representations, multiple feature extraction techniques will be employed:

- **Bag of Words (BoW):** To represent text using frequency of Bangla vocabulary terms.
- **TF-IDF:** To capture the importance of words based on their frequency across documents.
- **Word Embeddings:** Techniques such as Word2Vec and FastText will be implemented to capture semantic relationships between Bangla words.

3.3 Model Implementation

To perform multi-class sentiment classification, six models are implemented across three tiers, enabling a structured comparison from classical to state-of-the-art approaches.

Tier 1 – Classical Machine Learning Models

- **Logistic Regression:** A strong linear baseline trained on TF-IDF features, effective for multi-class text classification tasks.
- **Support Vector Machine (SVM):** Implemented using a linear kernel on TF-IDF representations, known for high performance on sparse text data.
- **Naive Bayes:** A probabilistic classifier trained on Bag-of-Words features, computationally lightweight and well-suited for noisy social media text.

Tier 2 – Deep Learning Models

- **Convolutional Neural Network (CNN):** Captures local n-gram patterns through multi-size convolutional filters, initialized with pre-trained FastText embeddings.
- **Bidirectional LSTM (BiLSTM):** Reads sequences in both forward and backward directions to model long-range contextual dependencies in Bangla text, initialized with Word2Vec embeddings.

Tier 3 – Transformer Model

- **BanglaBERT:** A BERT-based model pre-trained exclusively on large-scale Bangla corpora, fine-tuned on the project dataset to leverage deep contextual and linguistic representations specific to the Bangla language.

This three-tier architecture allows a systematic comparison across feature types, model complexity and linguistic understanding, ultimately identifying the most effective approach for Bangla social media sentiment analysis.

3.4 Training

The dataset is divided into training and testing subsets using a standard split (e.g., 80/20). The training process includes:

- Fitting models on the training data.
- Performing hyperparameter tuning to optimize model performance.
- Comparing results across different models and feature representations.

3.5 Implementation Tools

The system will be developed using the following tools and libraries:

- **Python** as the primary programming language.
- **Scikit-learn** for classical machine learning models.
- **TensorFlow/Keras** for building and training the deep learning models.

- **Hugging Face** for advanced NLP support.
- **Gensim** for training Word2Vec and FastText word embedding models on the Bangla corpus.
- **PyTorch** as the backend framework for BanglaBERT fine-tuning and inference.

4. Evaluation & Ablation Study

4.1 Evaluation Metrics

Model performance is evaluated using standard classification metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Percentiles
- Confusion Matrix

4.2 Validation Strategy

To ensure robustness and generalizability of the model:

- A train-test split for initial evaluation.
- Cross-validation to reduce variance and provide a more reliable estimate of performance.

4.3 Ablation Study

We are planning to analyze the contribution of different components of the system:

1. Feature Comparison:
    - TF-IDF vs Word Embeddings.
2. Preprocessing Impact:
    - Model performance before and after stopword removal.
3. Classical Model Comparison:
    - Logistic Regression vs SVM vs Naive Bayes under identical feature and preprocessing conditions to identify the strongest classical baseline.
4. Deep Learning Model Comparison
    - CNN vs BiLSTM, compared under the same embedding and preprocessing settings to evaluate the effect of model architecture on performance.
5. Tier-wise Progression Analysis
    - Classical ML vs Deep Learning vs Transformer (BanglaBERT), to demonstrate the performance gain achieved at each tier of model complexity.
6. Data Source Contribution:
    - Performance using only self-collected data vs combined dataset.

5. Gantt Chart

The proposed four-week project timeline is detailed below.

Week	Key Focus	Tasks
Week 1	Data Acquisition & Preprocessing	▸ Collect and manually label self-collected data and incorporate external datasets. ▸ Develop and apply the Data Preprocessing pipeline (removal of noise, stopword elimination, tokenization).
Week 2	Feature Extraction & Baseline Training	▸ Implement Feature Extraction techniques (BoW, TF-IDF, Word Embeddings). ▸ Implement and train initial classification models. ▸ Perform hyperparameter tuning.
Week 3	Comprehensive Evaluation & Ablations	▸ Finalize model training using cross-validation. ▸ Execute the full Ablation Study. ▸ Calculate all Evaluation Metrics.
Week 4	Analysis & Reporting	▸ Analyze results and generate all required graphs. ▸ Write the final project report. ▸ Final evaluation and determination of the optimal approach.

6. Expected Outcome

The expected outcomes of this project are as follows:

- Development of a functional and reliable sentiment classification system for Bangla social media text.
- Identification of the most effective combination of feature extraction techniques and classification models.
- Analysis of the impact of combining self-collected data with externally sourced datasets on model performance and generalization.
- Determination of an optimal approach for Bangla sentiment analysis, particularly for real-world social media text.